



## WRITTEN ASSIGNMENT

University of Applied Science - Online

Study-branch: MSc. Artificial Intelligence (120 ECTS)

### **Advanced Statistics Workbook (DLMDSAS01)**

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I List of Figures

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### III Abbreviations

|              |                                                     |
|--------------|-----------------------------------------------------|
| <b>AFL</b>   | American Fuzzy Lop                                  |
| <b>API</b>   | Application Programming Interface                   |
| <b>BIOS</b>  | Basic Input/Output System                           |
| <b>Brick</b> | Binary Run-time Integer Based Vulnerability Checker |
| <b>CaaS</b>  | Container as a Service                              |
| <b>CAB</b>   | Change Advisory Board                               |
| <b>CE</b>    | Community Edition                                   |
| <b>CI</b>    | Continuous Integration                              |
| <b>CLI</b>   | Command Line Interface                              |
| <b>CNCF</b>  | Cloud Native Computing Foundation                   |
| <b>CRED</b>  | C Range Error Detector                              |
| <b>Dev</b>   | Development, the development team                   |

# 1 Latex

## 1.1 Tools

MiKTeX: <https://miktex.org/download> TeXLive: <http://tug.org/texlive/> (or alternative LaTeX-systems).

A good editor is essential. Sometimes combined editors and compilers (e.g. TeXShop) can be really productive. Make sure you learn the use of synchronized navigation then.

A vector graphic is one where strokes remain strokes even at the highest resolutions: e.g. the Figure 2 or the Logo on the Titelblatt (notice: you can click from here to there). Many tools generate vector-graphics for plots from any data-set. E.g. Plotly (with the use of the Browser-Print), Matplotlib or even OpenOffice, LibreOffice or MS-Excel.

## 1.2 Literature References

Here is an example of a reference with a page-number: Dueck (2016, p. 6)

## 1.3 Pictures

## Task 1: Basic Probabilities and Visualizations (1)

$$\xi_1=3$$

$$\xi_2=0.61$$

A geometric distribution counting the number of Bernoulli trials with  $p = \xi_2$  until it succeeds.

```
#import libraries
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import geom
```

### Step 1:

Define Model

Define  $X \sim \text{Geometric}(p)$

and state the Probability Mass Function (PMF):  $P(X=k) = (1 - \xi_2)^{k-1} \xi_2$ .

```
# Define parameters based on xi_2
prob_success=0.61
prob_faliure=1-prob_success
# Initialize the Geometric distribution object (k >= 1)
dist = geom(prob_success)
```

### Step 2:

Compute  $E[X]$ , using the formula  $E[X] = 1/p$  for  $k \geq 1$  (Unit: 2.1, P. 48)

$$E[X] = \frac{1}{0.61} \approx 1.6393$$

```
E_X = 1 / prob_success
print(f"E[X]: {E_X:.4f}")
E[X]: 1.6393
```

### Step 3:

Compute the median  $M$ : find the smallest integer  $k$  such that the CDF satisfies  $P(X \leq k) \geq 0.5$ .  
Use  $P(X \leq k) = 1 - (1 - p)^k$ .

\* (Unit 2: 2.1, P. 50) \*

We solve for the smallest integer  $k$  such that:

$$(1 - 0.61)^k \leq 0.5$$

$$(0.39)^k \leq 0.5$$

$$k \geq \frac{\ln(0.5)}{\ln(0.39)} \approx 0.736$$

The smallest integer satisfying this is  $k=1$ .

```
# The median is found using the inverse CDF (Percent Point Function,
ppf)
M = dist.ppf(0.5)
print(f"Median M = {M:.0f}")

Median M = 1
```

#### Step 4:

Determine the upper limit  $K$  by finding the smallest integer  $k$  for which  $P(X=k) < 0.005$ . Explicitly cite the proof steps.

```
threshold = 0.005
k_values = []
# Iterate k starting from 1
for k in range(1, 15):
    prob = dist.pmf(k)
    print(f"P(X={k}) = {prob:.6f}, P(X=k) = {(prob_faliure**(k-1))*
(prob_success)} Condition P(X=k) < 0.005$? {prob < threshold}")
    if prob < threshold:
        # The range ends at k-1, since k itself is below the
        threshold.
        break
    k_values.append(k)

probabilities = dist.pmf(k_values)
K_max = max(k_values)
print(f"The visualization range is k = 1 to {K_max}")

P(X=1) = 0.610000, P(X=k) = 0.61 Condition P(X=k) < 0.005$? False
P(X=2) = 0.237900, P(X=k) = 0.2379 Condition P(X=k) < 0.005$? False
P(X=3) = 0.092781, P(X=k) = 0.092781 Condition P(X=k) < 0.005$? False
P(X=4) = 0.036185, P(X=k) = 0.03618459 Condition P(X=k) < 0.005$? False
P(X=5) = 0.014112, P(X=k) = 0.014111990100000003 Condition P(X=k) <
0.005$? False
P(X=6) = 0.005504, P(X=k) = 0.0055036761390000001 Condition P(X=k) <
0.005$? False
P(X=7) = 0.002146, P(X=k) = 0.0021464336942100004 Condition P(X=k)
< 0.005$? True
The visualization range is k = 1 to 6
```



### Step 5:

Generate the graphic (bar plot/discrete histogram) for  $k=1$  up to  $K$ . Include the scale. Present the calculated  $E[X]$  and  $M$  within the graphic.

We generate a bar plot to visualize the discrete probability mass function (PMF)

\$\*\_{{(Unit 5: 5.5, P. 139)}}\$

The graphic must include the scale of the axes and clearly display the calculated Expectation and Median

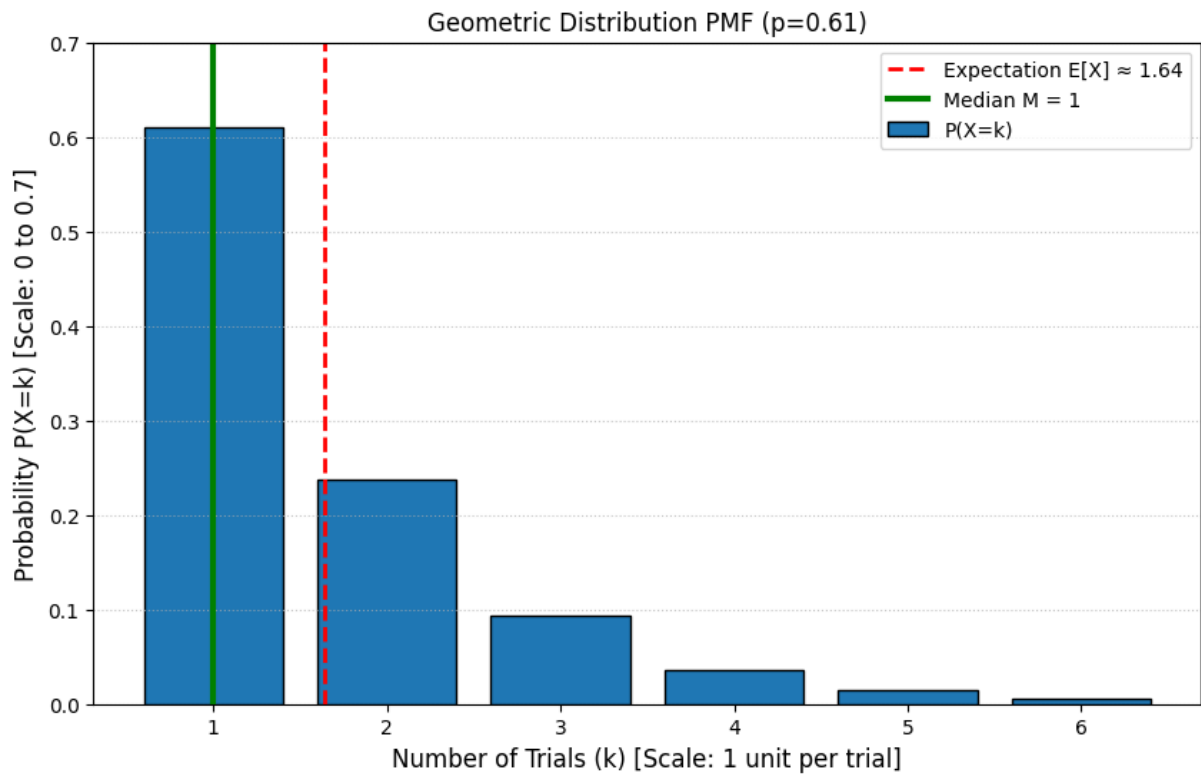
```
plt.figure(figsize=(10, 6))

# Plotting the PMF using a bar plot
plt.bar(k_values, probabilities, color='#1f77b4', edgecolor='black',
        label='P(X=k)')

# Mark Expectation
plt.axvline(E_X, color='red', linestyle='--', linewidth=2,
            label=f'Expectation E[X] ≈ {E_X:.2f}')

# Mark Median
plt.axvline(M, color='green', linestyle='--', linewidth=3,
            label=f'Median M = {M:.0f}')

# Formatting and Scale (required by assignment)
plt.title(f'Geometric Distribution PMF (p={prob_success})')
plt.xlabel('Number of Trials (k) [Scale: 1 unit per trial]',
           fontsize=12)
plt.ylabel('Probability P(X=k) [Scale: 0 to 0.7]', fontsize=12)
plt.xticks(k_values)
plt.ylim(0, 0.7) # Explicitly setting Y scale for clarity
plt.grid(axis='y', linestyle=':', alpha=0.7)
plt.legend()
plt.show()
```



## Justification of Tool Usage:

We utilize Python's standard statistical (scipy) and plotting (matplotlib) libraries.

These tools are widely accepted and trusted for producing accurate numerical results and visualizations in scientific computation.

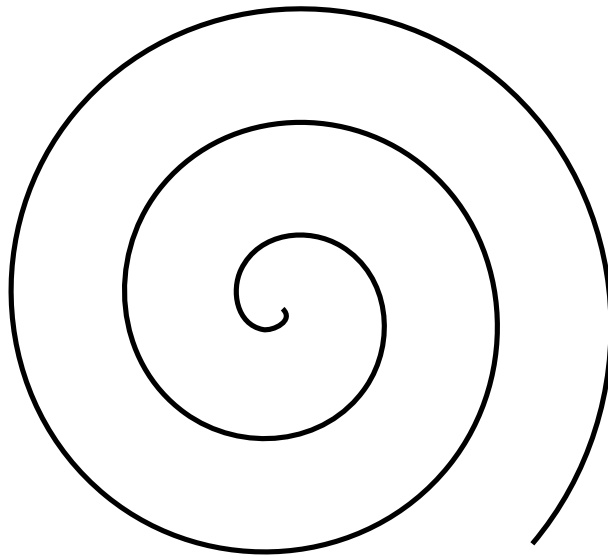


Figure 2: A spiral... smooth vector-based with a clean parametrisation!  
Nothing to do with Gage (2018)

## 1.5 Tables

| “ Industrial era ” | “Jobs ”        | “ Wanted: Upgrade”      |
|--------------------|----------------|-------------------------|
| Parts exchanger    | Fitter         | mecatronics special-ist |
| eShop              | reseller       | “ Client-suggester”     |
| “ Coding-guru”     | Softwaredesign | Whole-life designer     |
| JA! Gut & Günstig  | brand-names    | “ Life-Style Feeling”   |
| Internetbanking    | Bank clerk     | Customer adviser        |
| Robots             | Specialist     | Machine supervisor      |
| Bush               | Gardener       | Nature-sculptor         |
| Painting           | Painter        | Interior Design         |

Table 1: Downgrade and Upgrade of job denominations  
Dueck (2016)

## 1.6 Listes

- one
- twoi
- threei

1. first
2. second
3. third

## 1.7 Formulæ

A formula can be written inline, e.g. as  $\frac{d}{dx}\arctg(x) = \frac{1}{1+x^2}$  or, in centered math:

$$\frac{d}{dx}\arctg(x) = \frac{1}{1+x^2} \quad (1.1)$$

Notice that formulæ that are centered start bigger (technically, they start in `\displaystyle`) than they start inline (technically, they start in `\textstyle` all subsequent reductions, e.g. an exponent, goes to `\scriptstyle` then `\scriptscriptstyle`). Indeed a best effort is made so that inline formulæ do not change the line height which would bother the eye of a reader.

Formulæ can be given a number and a label. Numbering happens automatically with `\begin{equation}` and `\end{equation}` and can be avoided if enclosing the formula between `\[` and `\]`. If using the `\label` macro inside, you can refer automatically to this equation using `\ref{label}`. E.g. Thanks to equation 1.1 one dare say that:

$$\int_0^t \frac{1}{1+x^2} dx = \arctan(t) \quad (1.2)$$

## 1.8 Tools and Code

Many users of this template will want to include some code.

The simplest way to do so is to use the `\verb` macro which is followed by a sign, then some code, then the sign again to close. This is the inline version which works as in:

As we could calculate with `\textcite{Wolfram_Alpha}` using  
`\verb_integrate 1 / pi e ^ (t/pi) from zero to infinity_.`

which yields:

As we could calculate with Wolfram Inc. (2021) using `integrate 1 / pi e ^ (t/pi) from zero to infinity.`  
The multiline version of this is called `\begin{verbatim}` and finishes with `\end{verbatim}`.

## 1.9 Citation examples

Monography (GmbH, 2025, p. 22)

Collection (Müller & Mustermann, 2019)

Article (Schülers, 2021)

## 1.10 Task 1

## Bibliography

- Dueck, G. (2016). Zwischen industrialisierendem Downgrade und notwendigem Upgrade/Empowerment [auf dem AKAC-Kongress].
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- Müller, P., & Mustermann, M. (2019). Artikel in Sammelband [Sammelband für IUBH]. Verlag.
- Schülers. (2021). Zeitschriftenartikel [In: Frankfurter Allgemeine Zeitung], 10. p. 10 –14.
- Wolfram Inc. (2021). Wolfram Alpha [Accessed: 10.06.2021]. <https://wolframalpha.com>